Math 428:
Algorithms and Numerical Solutions
of
Differential Equations

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Part 0: Course logistics

Course overview and syllabus

- Description
- Objectives
- Resources
  → Time, textbook, office hours, etc.
- Grading
- Exams, homework, projects, etc.
- A word on student conduct...

Course overview and syllabus

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  → Straight scale
- Exams, homework, projects, etc.
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Course overview and syllabus

- Description
  ... Scientific computation can be studied abstractly, and it can be practiced by professionals to solve problems. This course emphasizes both in equal parts. ...
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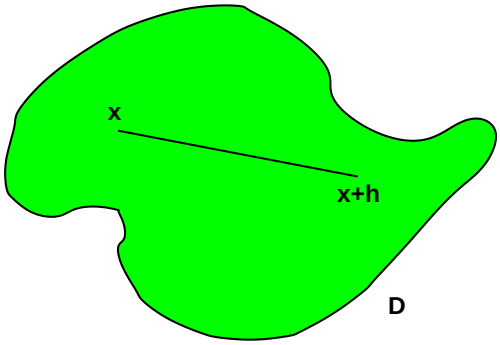

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Taylor's theorem (n-D)

For example, consider a first order approximation in 3 dimensions.

$$f(\vec{x}+\vec{h}) =$$
$$f(\vec{x}) + \frac{\partial f}{\partial x_1}(\vec{x})h_1 + \frac{\partial f}{\partial x_2}(\vec{x})h_2 + \frac{\partial f}{\partial x_3}(\vec{x})h_3 +$$
$$\frac{1}{2}\frac{\partial^2 f}{\partial x_1^2}(\vec{\xi})h_1^2 + \frac{1}{2}\frac{\partial^2 f}{\partial x_2^2}(\vec{\xi})h_2^2 + \frac{1}{2}\frac{\partial^2 f}{\partial x_3^2}(\vec{\xi})h_3^2 +$$
$$\frac{\partial^2 f}{\partial x_1 \partial x_2}(\vec{\xi})h_1h_2 + \frac{\partial^2 f}{\partial x_2 \partial x_3}(\vec{\xi})h_2h_3 + \frac{\partial^2 f}{\partial x_1 \partial x_3}(\vec{\xi})h_1h_3$$

Part 1: Essentials

Part 2: Numerical quadrature

Essentials - Taylor's theorem (1D)

Taylor's theorem is the first (and often best) tool to locally analyze an approximation. (See KC, pp 6-9 for a primer.)

Theorem (Taylor):
If $f \in \mathrm{C}^{n+1}([a,b])$, then for any points x and $x+h$ in the interval $[a,b]$,

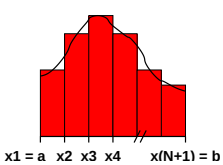
$$f(x+h) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x)h^k + \frac{1}{(n+1)!} f^{(n+1)}(\xi)h^{n+1}$$

where ξ lies between x and $x+h$. The latter term, called the remainder, often corresponds to the error in some approximate scheme.

A naive look at numerical quadrature

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$. Remember, midpoint rule from calculus...

$$\int_a^b f(x)dx \approx \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right)h$$



Taylor's theorem (n-D)

Theorem (Taylor):
If $f \in \mathrm{C}^{n+1}(D)$, then for any points $\vec{x} = [x_1, x_2, \ldots x_m]^T$ and $\vec{x}+\vec{h}$ in a region $D \in \mathrm{R}^m$ containing the line segment from $\vec{x}$ to $\vec{x}+\vec{h}$,

$$f(\vec{x}+\vec{h}) =$$
$$\sum_{k=0}^{n} \frac{1}{k!} \sum_{|\vec{i}|=k} \frac{\partial^k f}{\partial x_1^{i_1} \partial x_2^{i_2} \ldots \partial x_m^{i_m}}(\vec{x}) h_1^{i_1} h_2^{i_2} \ldots h_m^{i_m} +$$
$$\frac{1}{(n+1)!} \sum_{|\vec{i}|=n+1} \frac{\partial^{n+1} f}{\partial x_1^{i_1} \partial x_2^{i_2} \ldots \partial x_m^{i_m}}(\vec{\xi}) h_1^{i_1} h_2^{i_2} \ldots h_m^{i_m}$$

where $\vec{\xi}$ lies between $\vec{x}$ and $\vec{x}+\vec{h}$, and $\vec{i}$ is a multi-index.

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$.
Remember, midpoint rule from calculus...
And, this is true for any odd function of a symmetric interval.

For instance... $$\int_{x_i}^{x_{i+1}} \left(x - \frac{x_{i+1}+x_i}{2}\right)^3 dx = 0$$

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$.
Remember, midpoint rule from calculus...

$$\int_a^b f(x)dx \approx \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right) h$$

$$\int_a^b f(x)dx = \sum_{i=1}^{N} \int_{x_i}^{x_{i+1}} f(x)dx$$

A naive look at numerical quadrature

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$.
Remember, midpoint rule from calculus...

$$\int_a^b f(x)dx = \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right) h +$$
$$\sum_{i=1}^{N} \int_{x_i}^{x_{i+1}} \frac{1}{2} f''(\xi_i) \left(x - \frac{x_{i+1}+x_i}{2}\right)^2 dx$$

Note: ξ_i depends upon x, the variable of integration.

A naive look at numerical quadrature

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$.
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$$\int_a^b f(x)dx \approx \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right) h$$

$$\int_a^b f(x)dx = \sum_{i=1}^{N} \int_{x_i}^{x_{i+1}} \left\{ f\left(\frac{x_{i+1}+x_i}{2}\right) + \right.$$
$$\left. f'\left(\frac{x_{i+1}+x_i}{2}\right)\left(x - \frac{x_{i+1}+x_i}{2}\right) + \frac{1}{2} f''(\xi_i) \left(x - \frac{x_{i+1}+x_i}{2}\right)^2 \right\} dx$$

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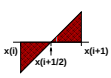
Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$.
Remember, midpoint rule from calculus...

$$\left| \int_a^b f(x)dx - \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right) h \right| =$$
$$\left| \sum_{i=1}^{N} \int_{x_i}^{x_{i+1}} \frac{1}{2} f''(\xi_i) \left(x - \frac{x_{i+1}+x_i}{2}\right)^2 dx \right|$$

A naive look at numerical quadrature

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$.
Remember, midpoint rule from calculus...
Let's remember why that red term is zero. This will be a very common theme in this course.

$$\int_{x_i}^{x_{i+1}} f'\left(\frac{x_{i+1}+x_i}{2}\right)\left(x - \frac{x_{i+1}+x_i}{2}\right) dx =$$
$$f'\left(\frac{x_{i+1}+x_i}{2}\right) \int_{x_i}^{x_{i+1}} \left(x - \frac{x_{i+1}+x_i}{2}\right) dx =$$
$$f'\left(\frac{x_{i+1}+x_i}{2}\right) \cdot 0 = 0$$



Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$. Remember, midpoint rule from calculus...

$$\left|\int_a^b f(x)dx - \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right)h\right| \le \frac{Mh^2(b-a)}{24} = \frac{M(b-a)^3}{24N^2}$$

where $|f''(\xi)| \le M$ for all $\xi \in [a,b]$.
Thus, this is a crude, $O(N^2)$, method.

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$. Remember, midpoint rule from calculus...

$$\left|\int_a^b f(x)dx - \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right)h\right| \le \frac{1}{2}\sum_{i=1}^{N}\int_{x_i}^{x_{i+1}}\left|f''(\xi_i)\left(x-\frac{x_{i+1}+x_i}{2}\right)^2\right|dx$$

This is nothing more than the *triangle inequality* applied to the sum and the integral.

An example using Matlab

We would like to accurately compute

$$\int_1^2 \frac{1}{x}dx.$$

Define the function in Matlab (in a simple file called `midptf.m`).

```
function y = midptf(x)
  y = 1/x;
```

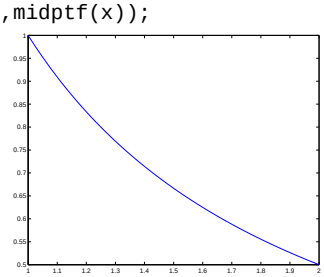

A naive look at numerical quadrature

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$. Remember, midpoint rule from calculus...

$$\left|\int_a^b f(x)dx - \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right)h\right| \le \frac{1}{2}\sum_{i=1}^{N}\int_{x_i}^{x_{i+1}}\left|f''(\xi_i)\right|\left|\left(x-\frac{x_{i+1}+x_i}{2}\right)^2\right|dx$$

An example using Matlab (cont'd)

```
>> x = linspace(1,2,200);
>> plot(x,midptf(x));
```




```
Looks good!  Let's try the midpoint rule.
```

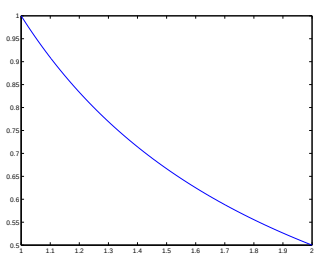

A naive look at numerical quadrature

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$. Remember, midpoint rule from calculus...

$$\left|\int_a^b f(x)dx - \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right)h\right| \le \frac{M}{2}\sum_{i=1}^{N}\int_{x_i}^{x_{i+1}}\left(x-\frac{x_{i+1}+x_i}{2}\right)^2 dx$$

where $|f''(\xi)| \le M$ for all $\xi \in [a,b]$.

An example using Matlab (cont'd)



Remember, we expect to pick up one digit of accuracy for every factor of 10 decrease in h^2. We know the exact answer, $\ln 2$, but in practice we would not. However, by choosing h really, really small, we would have a good approximation that we could use for a *refinement study.*

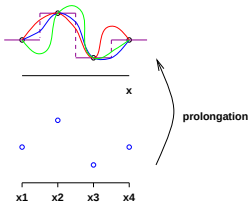
A naive look at numerical quadrature

Suppose we would like to compute the area under the curve of some function $f(x)$ on the interval $[a,b]$. Remember, midpoint rule from calculus...

$$\left|\int_a^b f(x)dx - \sum_{i=1}^{N} f\left(\frac{x_{i+1}+x_i}{2}\right)h\right| \le \frac{M}{2}\sum_{i=1}^{N}\frac{h^3}{12}$$

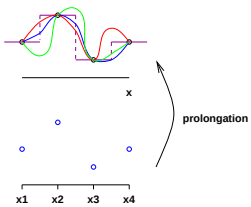
where $|f''(\xi)| \le M$ for all $\xi \in [a,b]$. Remember, $h = \frac{b-a}{N}$ or $N = \frac{b-a}{h}$.

Consider that there are really two functions. One function $f(x)$ is a function of a continuous variable $x \in [a, b]$. The other, $\widehat{f}(\widehat{x})$, is a discrete function of discrete variable, $\widehat{x} \in \{x_1, x_2, \ldots, x_n\}$. Moving from one space to another is called *restriction* and *prolongation*.



Notice that prolongation is not so easy to define.

Another way to look the midpoint rule



Notice that prolongation is not so easy to define.
The midpoint rule uses piecewise constant functions to prolong the discrete function $\widehat{f}$. The resulting integral is the exact integral of this prolongation.

Making better methods...

Now that we understand the midpoint rule so well, let's try to improve it using our error analysis. If M_N is the midpoint quadrature using N panels or intervals, we know how to estimate the error over one interval.

$$\left| \int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h \right| \leq \frac{|f''(\xi)|h^3}{24}$$

where $\xi \in [x_i, x_{i+1}]$ and $x_{i+\frac{1}{2}} = \frac{x_i + x_{i+1}}{2}$. But, with an eye toward improvement, let's not bound things right away. Rather, let's keep an extra terms from the Taylor series.

Making better methods...

Now that we understand the midpoint rule so well, let's try to improve it using our error analysis. If M_N is the midpoint quadrature using N panels or intervals, we know how to estimate the error over one interval.

$$\int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h =$$
$$\int_{x_i}^{x_{i+1}} \left[f'(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}}) + \frac{1}{2} f''(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}})^2 + \right.$$
$$\left. \frac{1}{3!} f'''(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}})^3 + \frac{1}{4!} f^{(iv)}(\xi)(x - x_{i+\frac{1}{2}})^4 \right] dx$$

The two red terms are zero (why?).

N	Error
5	0.0012393

```
>> x = linspace(1,2,6);
>> m5 = sum(midptf((x(1:5)+x(2:6))/2))* ...
(x(2)-x(1));
>> log(2)-m5

ans =

     1.239294844010197e-03
```


An example using Matlab (cont'd)

N	Error
5	0.0012393
10	0.0003118

```
>> x = linspace(1,2,11);
>> m10 = sum(midptf((x(1:10)+x(2:11))/2))* ...
(x(2)-x(1));
>> log(2)-m10

ans =

     3.118201499844320e-04
```


An example using Matlab (cont'd)

N	Error
5	0.0012393
10	0.0003118
20	0.0000781

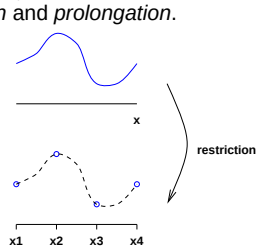
```
>> x = linspace(1,2,21);
>> m20 = sum(midptf((x(1:20)+x(2:21))/2))* ...
(x(2)-x(1));
>> log(2) - m20

ans =

     7.808233435779854e-05
```


Another way to look the midpoint rule

Consider that there are really two functions. One function $f(x)$ is a function of a continuous variable $x \in [a, b]$. The other, $\widehat{f}(\widehat{x})$, is a discrete function of discrete variable, $\widehat{x} \in \{x_1, x_2, \ldots, x_n\}$. Moving from one space to another is called *restriction* and *prolongation*.



Restriction can be defined explicitly in many settings.

Definition (Big O):
The expression $O(h^p)$ (said "big oh of h to the p") is defined as follows.

$$f(h) = g(h) + O(h^p)$$

means that

$$|f(h) - g(h)| \leq Ch^p$$

for all h and some fixed p where the constant C does not depend on h.

Example (Big O example 1):
From Taylor's Theorem, we know that

$$f(x+h) = f(x) + O(h).$$

Now that we understand the midpoint rule so well, let's try to improve it using our error analysis. If M_N is the midpoint quadrature using N panels or intervals, we know how to estimate the error over one interval.

$$\int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h =$$
$$\int_{x_i}^{x_{i+1}} \left[\frac{1}{2}f''(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}})^2 + \frac{1}{4!}f^{(iv)}(\xi)(x - x_{i+\frac{1}{2}})^4\right] dx$$

Remember that the ξ depends upon the variable of integration, x, so we cannot evaluate the integral of the last term.

Aside: Big O notation

Definition (Big O):
The expression $O(h^p)$ (said "big oh of h to the p") is defined as follows.

$$f(h) = g(h) + O(h^p)$$

means that

$$|f(h) - g(h)| \leq Ch^p$$

for all h and some fixed p where the constant C does not depend on h.

Example (Big O example 2):
From Taylor's Theorem, we also know that

$$f(x+h) = f(x) + f'(x)h + O(h^2).$$

Making better methods...

Now that we understand the midpoint rule so well, let's try to improve it using our error analysis. If M_N is the midpoint quadrature using N panels or intervals, we know how to estimate the error over one interval.

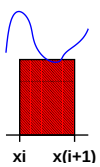
$$\int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h =$$
$$\frac{1}{24}f''(x_{i+\frac{1}{2}})h^3 + \frac{1}{4!}\int_{x_i}^{x_{i+1}} f^{(iv)}(\xi)(x - x_{i+\frac{1}{2}})^4 dx$$

Remember that the ξ depends upon the variable of integration, x. Here, we must stop now that we have identified an specific error term that we wish to eliminate.

Making better methods (cont'd)...

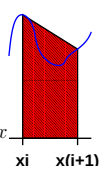
Midpoint rule leading order error:

$$\int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h =$$
$$\frac{1}{24}f''(x_{i+\frac{1}{2}})h^3 + C\int_{x_i}^{x_{i+1}} f^{(iv)}(\xi)(x - x_{i+\frac{1}{2}})^4 dx$$



It can be shown that for trapezoid rule:

$$\int_{x_i}^{x_{i+1}} f(x)dx - \left[\frac{1}{2}f(x_i) + \frac{1}{2}f(x_{i+1})\right] h =$$
$$-\frac{1}{12}f''(x_{i+\frac{1}{2}})h^3 + D\int_{x_i}^{x_{i+1}} f^{(iv)}(\xi)(x - x_{i+\frac{1}{2}})^4 dx$$



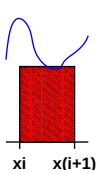
Aside: Big O notation

Often in mathematics, we are interested in the general behavior of a quantity with respect to a variable, but the specifics to not matter. For instance, the error terms $5h^2$ and $10h^2$ mean almost the same thing. Though the first term is smaller than the second, both errors refer to quadratic convergence. We introduce a new notation $O(h^2)$ to describe both terms.

Making better methods (cont'd)...

Midpoint rule leading order error:

$$\int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h =$$
$$\frac{1}{24}f''(x_{i+\frac{1}{2}})h^3 + O(h^5)$$



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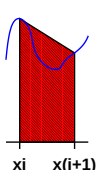
$$-\frac{1}{12}f''(x_{i+\frac{1}{2}})h^3 + O(h^5)$$



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means that

$$|f(h) - g(h)| \leq Ch^p$$

for all h and some fixed p where the constant C does not depend on h.

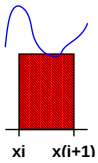
Or, if M_N is the midpoint rule, T_N is the trapezoid rule and S_N is Simpson's rule, then

$$S_N = \frac{2}{3}M_N + \frac{1}{3}T_N.$$

Note: Simpson's rule is often written with *double panels*, treating the midpoint and the endpoints of each panel as a separate mesh point. This effectively doubles N.

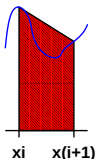
$$\int_a^b f(x)dx \approx$$
$$\frac{h}{6}\left[f(x_1) + 4f(x_2) + 2f(x_3) + 4f(x_4) + 2f(x_5) + \ldots + f(x_{N+1})\right]$$

Midpoint rule leading order error:



$$\int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h =$$
$$\frac{1}{24}f''(x_{i+\frac{1}{2}})h^3 + O(h^5)$$

It can be shown that for trapezoid rule:



$$\int_{x_i}^{x_{i+1}} f(x)dx - \left[\frac{1}{2}f(x_i) + \frac{1}{2}f(x_{i+1})\right]h =$$
$$-\frac{1}{12}f''(x_{i+\frac{1}{2}})h^3 + O(h^5)$$

How can we use this information to develop a better method?

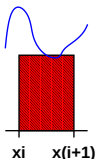
A matlab interlude...

Consider that nasty old integral

$$\int_{\frac{1}{10}}^{1} \frac{\cos\left(\frac{\ln x}{x}\right)}{x}dx.$$

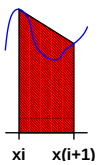
Making better methods (cont'd)...

Midpoint rule leading order error:



$$\int_{x_i}^{x_{i+1}} f(x)dx - f(x_{i+\frac{1}{2}})h =$$
$$\frac{1}{24}f''(x_{i+\frac{1}{2}})h^3 + O(h^5)$$

It can be shown that for trapezoid rule:



$$\int_{x_i}^{x_{i+1}} f(x)dx - \left[\frac{1}{2}f(x_i) + \frac{1}{2}f(x_{i+1})\right]h =$$
$$-\frac{1}{12}f''(x_{i+\frac{1}{2}})h^3 + O(h^5)$$

A matlab interlude...

Consider that nasty old integral

$$\int_{\frac{1}{10}}^{1} \frac{\cos\left(\frac{\ln x}{x}\right)}{x}dx.$$

N	M_N	$M_{2N} - M_N$	T_N	$T_{2N} - T_N$	S_N	$S_{2N} - S_N$
100	2.8664e-01	8.8986e-03	3.1909e-01	-1.6223e-03	2.9746e-01	5.2458e-04
200	2.9554e-01	1.8820e-03	3.0286e-01	-3.6624e-03	2.9798e-01	3.3855e-05
400	2.9742e-01	4.4824e-04	2.9920e-01	-8.9022e-04	2.9801e-01	2.0871e-06

Simpson's Rule

Eliminate the leading error terms from midpoint and trapezoidal rule because they are two comparable $O(h^2)$ methods.

$$\frac{1}{3}\left\{3\int_{x_i}^{x_{i+1}} f(x)dx - 2f(x_{i+\frac{1}{2}})h - \left[\frac{1}{2}f(x_i) + \frac{1}{2}f(x_{i+1})\right]h\right\} =$$
$$\frac{1}{3}\left[2\frac{1}{24}f''(x_{i+\frac{1}{2}})h^3 - \frac{1}{12}f''(x_{i+\frac{1}{2}})h^3\right] + O(h^5)$$

Simpson's rule: Another perspective.

One can also derive Simpson's rule by using three points per panel to interpolate $f(x)$ with a parabola, $p(x)$. The Simpson's rule approximation is the area under the parabola.

$$p(x) = f(x_i)\frac{(x - x_{i+\frac{1}{2}})(x - x_{i+1})}{(x_i - x_{i+\frac{1}{2}})(x_i - x_{i+1})} +$$
$$f(x_{i+\frac{1}{2}})\frac{(x - x_i)(x - x_{i+1})}{(x_{i+\frac{1}{2}} - x_i)(x_{i+\frac{1}{2}} - x_{i+1})} +$$
$$f(x_{i+1})\frac{(x - x_i)(x - x_{i+\frac{1}{2}})}{(x_{i+1} - x_i)(x_{i+1} - x_{i+\frac{1}{2}})}.$$

This is Newton's interpolating polynomial for these three data points. We shall be seeing more of these.

Simpson's Rule

Eliminate the leading error terms from midpoint and trapezoidal rule because they are two comparable $O(h^2)$ methods.

$$\int_{x_i}^{x_{i+1}} f(x)dx - \frac{2}{3}f(x_{i+\frac{1}{2}})h - \left[\frac{1}{6}f(x_i) + \frac{1}{6}f(x_{i+1})\right]h = O(h^5)$$

Or, if M_N is the midpoint rule, T_N is the trapezoid rule and S_N is Simpson's rule, then

$$S_N = \frac{2}{3}M_N + \frac{1}{3}T_N.$$

The *global* accuracy of this quadrature is $O(h^4)$ because we must add up $N = O(h^{-1})$ errors.

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

Now, let's see what the midpoint rule does and compare it to the exact integral.

$$M_N = h \sum_{i=1}^{N} p(x_{i+\frac{1}{2}})$$

where $h = \frac{L}{N}$ and $x_{i+\frac{1}{2}} = h\left(i - \frac{1}{2}\right) = \frac{L}{N}\left(i - \frac{1}{2}\right)$.

Assuming $f(x)$ is continuously differentiable enough times, integrating $f(x)$ is the same as integrating its Taylor series. Thus, you can judge a quadrature by how well it integrates polynomials.

- The midpoint rule *exactly* integrates constant functions and linear functions $f(x) = c_0 + c_1 x$, so we conclude that its accuracy is *at least* $O(h^2)$. We have seen that this is true.
- The same reasoning can be applied to the trapezoid rule.
- Simpson's rule will *exactly* integrate cubic polynomials, $f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$!

The downside of this approach is that it does not appear to tell you the exact form of the leading error term...

A cruder way to gauge accuracy.

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

$$\begin{aligned} M_N &= \frac{L}{N} \sum_{i=1}^{N} \left[c_0 + c_1 \frac{L}{N}\left(i - \frac{1}{2}\right) + \frac{c_2}{2}\left(\frac{L}{N}\right)^2 \left(i - \frac{1}{2}\right)^2 \right] \\ &= \frac{L}{N}\left[\sum_{i=1}^{N} c_0 + c_1 \frac{L}{N} \sum_{i=1}^{N}\left(i - \frac{1}{2}\right) + \frac{c_2}{2}\left(\frac{L}{N}\right)^2 \sum_{i=1}^{N}\left(i - \frac{1}{2}\right)^2 \right] \\ &= \frac{L}{N}\left[c_0 N + c_1 \frac{L}{N} \sum_{i=1}^{N} i - c_1 \frac{L}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^2 \sum_{i=1}^{N}\left(i^2 - i + \frac{1}{4}\right) \right] \end{aligned}$$

A cruder way to gauge accuracy.

In fact, this is not true! For instance, we can study the midpoint rule by testing it against a quadratic.

$$p(x) = c_0 + c_1 x + \frac{1}{2} c_2 x^2.$$

Written this way, c_2, is the second derivative of my little test polynomial.
Also, we can simplify the problem a bit because know that the specific endpoints, $[a, b]$, of the interval do not matter so much as the length of the interval. So, we shall let $L = b - a$, and integrate on $[0, L]$.
The exact integral is easy to find.

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}.$$

A cruder way to gauge accuracy.

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

$$M_N = \frac{L}{N}\left[c_0 N + c_1 \frac{L}{N} \sum_{i=1}^{N} i - c_1 \frac{L}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^2 \sum_{i=1}^{N}\left(i^2 - i + \frac{1}{4}\right) \right]$$

Recall,

$$\sum_{i=1}^{N} i = \frac{N(N+1)}{2}.$$

Maple knows all these identities by the way...

A cruder way to gauge accuracy.

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

Now, let's see what the midpoint rule does and compare it to the exact integral.

$$M_N = h \sum_{i=1}^{N} p(\ x_{i+\frac{1}{2}}\ )$$

where $h = \frac{L}{N}$ and $x_{i+\frac{1}{2}} = h\left(i - \frac{1}{2}\right) = \frac{L}{N}\left(i - \frac{1}{2}\right)$.

A cruder way to gauge accuracy.

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

$$M_N = \frac{L}{N}\left[c_0 N + c_1 \frac{L}{N} \frac{N(N+1)}{2} - c_1 \frac{L}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^2 \sum_{i=1}^{N}\left(i^2 - i + \frac{1}{4}\right) \right]$$

A cruder way to gauge accuracy.

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

Now, let's see what the midpoint rule does and compare it to the exact integral.

$$M_N = h \sum_{i=1}^{N} p(x_{i+\frac{1}{2}})$$

where $h = \frac{L}{N}$ and $x_{i+\frac{1}{2}} = h\left(i - \frac{1}{2}\right) = \frac{L}{N}\left(i - \frac{1}{2}\right)$.

We have discussed two methods of analyzing the accuracy of a given quadrature, I_N.

- Taylor expansion method.
  - Begin with $\int_{x_i}^{x_{i+1}} f(x)dx - I_N$.
  - Use Taylor's theorem to expand $f(x)$ to reproduce the quadrature at zeroth order.
  - Integrate the remaining terms of the Taylor's series until a nonzero term emerges.
  - This nonzero term is the error over one panel.
  - Sum the error over all panels to determine to total error.
- Polynomial test function method.

$$\begin{aligned}
\sum_{i=1}^{N} i &= \frac{N(N+1)}{2} \\
\sum_{i=1}^{N} i^2 &= \frac{N(N+1)(2N+1)}{6} \\
\sum_{i=1}^{N} i^3 &= \frac{N^2(N+1)^2}{4} \\
\sum_{i=1}^{N} i^4 &= \frac{N(N+1)(2N+1)(3N^2+3N+1)}{30}
\end{aligned}$$

Again, Maple can find these for you. Just ask:

```
> simplify(sum(i^5,i=1..N));
```


Summary: Accuracy of quadratures

We have discussed two methods of analyzing the accuracy of a given quadrature, I_N.

- Taylor expansion method.
- Polynomial test function method.
  - Insert a test polynomial of order q, $p(x) = \sum_{k=0}^{q} \frac{c_k}{k!} x^k$ into the quadrature.
  - Compare the sum to the exact integral of the polynomial.
  - The first term that does not agree is the error. The leading coefficient, c_n, represents the dependence on the nth derivative of the function being integrated.

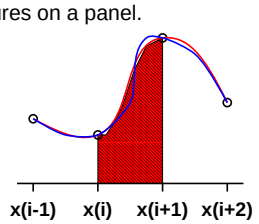
A cruder way (cont'd)

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

$$M_N = \frac{L}{N}\left[c_0 N + c_1 \frac{L}{N}\frac{N(N+1)}{2} - c_1 \frac{L}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^2 \sum_{i=1}^{N}\left(i^2 - i + \frac{1}{4}\right)\right]$$

Designing quadratures

So, it would seem that there are a lot of different ways to design quadratures on a panel.



So, we can interpolate a function using a lot of points inside a panel or we can use outside points to interpolate a function. No matter what we do, we must learn about polynomial interpolation.

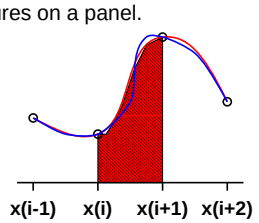
A cruder way (cont'd)

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

$$\begin{aligned}
M_N &= \frac{L}{N}\left[c_0 N + c_1 \frac{LN}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^2 \sum_{i=1}^{N}\left(i^2 - i + \frac{1}{4}\right)\right] \\
&= c_0 L + c_1 \frac{L^2}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^3 \sum_{i=1}^{N}\left(i^2 - i + \frac{1}{4}\right)
\end{aligned}$$

Designing quadratures

So, it would seem that there are a lot of different ways to design quadratures on a panel.



This quadrature would be called $Q_{3,1}$ because we are using a third order (cubic) quadrature (over four points), and we are integrating over one subinterval.

A cruder way (cont'd)

$$\int_0^L p(x)dx = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{6}$$

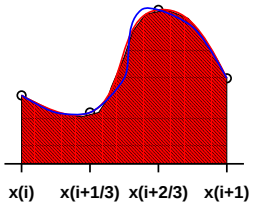
$$\begin{aligned}
M_N &= c_0 L + c_1 \frac{L^2}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^3 \sum_{i=1}^{N}\left(i^2 - i + \frac{1}{4}\right) \\
&= c_0 L + c_1 \frac{L^2}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^3 \left[\frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)}{2} + \frac{N}{4}\right] \\
&= c_0 L + c_1 \frac{L^2}{2} + \frac{c_2}{2}\left(\frac{L}{N}\right)^3 \frac{4N^3 - N}{12} \\
&= c_0 L + c_1 \frac{L^2}{2} + \frac{c_2 L^3}{6} - \frac{c_2 L^3}{24N^2}
\end{aligned}$$

Given a set of $n+1$ mesh points, we will discuss two ways to finding the interpolating polynomial of degree p.

- Lagrange polynomials: Easy to manipulate theoretically. Tougher to actually compute.
- Divided differences: Nasty to manipulate. Easier to compute mechanically.

Another method which is less practical uses Vandermonde matrices (see Ch. 6.1).

So, it would seem that there are a lot of different ways to design quadratures on a panel.



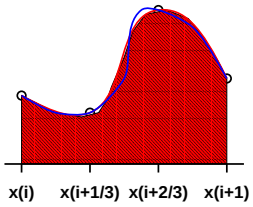
This quadrature would be called $Q_{3,3}$ because we are using a third order quadrature (over four points), and we are integrating over three subintervals.

Lagrange polynomials

Given a set of $n+1$ mesh points, there is a unique polynomial of degree n that passes each point $(x_i, f(x_i))$. One way to express this polynomial is to use Lagrange's interpolation formula.

Designing quadratures

So, it would seem that there are a lot of different ways to design quadratures on a panel.



Generally, $Q_{p,q}$, means we find a pth degree polynomial extending over $p+1$ points, and integrate it over the inner q subpanels. There is a whole "industry" dedicated to designing these quadratures, but we will focus on Newton-Cotes quadratures which is when $p = q$.

Lagrange polynomials

Given a set of $n+1$ mesh points, there is a unique polynomial of degree n that passes each point $(x_i, f(x_i))$. One way to express this polynomial is to use Lagrange's interpolation formula.

$$p(x) = \sum_{i=1}^{n+1} f(x_i) l_i(x)$$

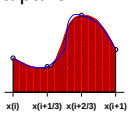
where

$$l_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

Notice, each $l_i(x)$ has degree n, and $p(x)$ passes through each data point.

Designing quadratures

So, it would seem that there are a lot of different ways to design quadratures on a panel.



For instance, $Q_{2,2}$ is Simpson's rule. This quadrature is results in a method,

$$\int_{x_i}^{x_{i+1}} f(x)dx \approx h\left[\frac{1}{8}f(x_i) + \frac{3}{8}f(x_{i+\frac{1}{3}}) + \frac{3}{8}f(x_{i+\frac{2}{3}}) + \frac{1}{8}f(x_{i+1})\right],$$

also known as "Simpson's $\frac{3}{8}$ rule."

Lagrange polynomials

Consider an interpolation over the data points $f(0) = 2$, $f(2) = 1$, $f(3) = -1$ and $f(4) = 1$.
For example,

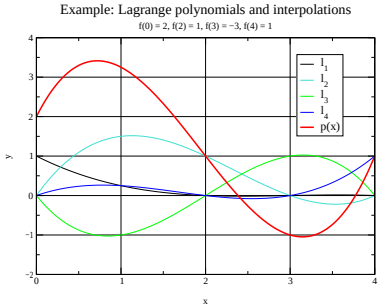
$$l_1 = \frac{(x-2)(x-3)(x-4)}{(-2)(-3)(-4)}$$

Designing quadratures

General comments, offered without proof, on Newton-Cotes quadratures.

- If n is even, $Q_{n,n}$ is $O(n^{n+3})$.
- If n is odd, $Q_{n,n}$ is $O(n^{n+2})$.

Consider an interpolation over the data points $f(0) = 2$, $f(2) = 1$, $f(3) = -1$ and $f(4) = 1$.



Lagrange polynomials

With remainder,

$$f(x) = \sum_{i=1}^{n+1} f(x_i) l_i(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=1}^{n+1} (x - x_i)$$

where

$$l_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

and ξ is somewhere on the interval $[x_1, x_{n+1}]$.

Yet another way to think about interpolating polynomials is using divided differences.
Each divided difference approximates a derivative on the mesh.

$$f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

This expression approximates the derivative of f on $[x_1, x_2]$.
Similarly, a second derivative can be approximated recursively.

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}.$$

This is a difference of differences.

Divided differences

So, we can build a table of these differences.

$$\begin{array}{llll}
x_1 & f[x_1] & & \\
 & & f[x_1, x_2] & \\
x_2 & f[x_2] & & f[x_1, x_2, x_3] \\
 & & f[x_2, x_3] & \\
x_3 & f[x_3] & &
\end{array}$$

And, we can use such tables to express the interpolating polynomial,

$$p(x) = f[x_1] + f[x_1, x_2](x - x_1) + f[x_1, x_2, x_3](x - x_1)(x - x_2).$$

See Chapter 6.2 for further discussion and an example.

Divided differences

Let's try the example (again): $f(0) = 2$, $f(2) = 1$, $f(3) = -1$ and $f(4) = 1$

$$\begin{array}{llllll}
0 & 2 & & & \\
 & & -1/2 & & \\
2 & 1 & & -1/2 & \\
 & & -2 & & 5/8 \\
3 & -1 & & 2 & \\
 & & 2 & & \\
4 & 1 & & &
\end{array}$$

And, we can use such tables to express the interpolating polynomial,

$$p(x) = 2 - \frac{1}{2}x - \frac{1}{2}x(x-2) + \frac{5}{8}x(x-2)(x-3).$$

Divided differences

Yet another way to think about interpolating polynomials is using divided differences.
Each divided difference approximates a derivative on the mesh.

$$f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

This expression approximates the derivative of f on $[x_1, x_2]$.
Remember,

$$\frac{df}{dx}(x_2) = \lim_{x_2 \to x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Gaussian quadrature

We have seen how one can design quadratures by choosing mesh points and determining the "weights" for each point based on interpolating polynomials. Thus, a general n-point quadrature over one panel has a form

$$I \approx h \sum_{i=1}^{n} \alpha_k f(x_{i_k}).$$

For instance, for the midpoint rule, $n = 1$, $x_{i_1} = x_{i+\frac{1}{2}}$, and $\alpha_1 = 1$. For the trapezoid rule, $n = 2$, $x_{i_1} = x_i$, $x_{i_2} = x_{i+1}$, $\alpha_1 = \frac{1}{2}$ and $\alpha_2 = \frac{1}{2}$. The key point is that the quadrature points are chosen, and the weights are computed.

Yet another way to think about interpolating polynomials is using divided differences.
Each divided difference approximates a derivative on the mesh.

$$f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

This expression approximates the derivative of f on $[x_1, x_2]$.
Similarly, a second derivative can be approximated recursively.

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}.$$

This is a difference of differences.

Gauss proposed choosing both the mesh points and the weights so that one would have more degrees of freedom and therefore higher accuracy. For instance, a two-point quadrature does not have to be the trapezoid rule. We can select x_{i_1} and x_{i_2} to make the quadrature more accurate. If I can choose the α's and the x_{i_k}'s, I should get $O(h^{2n})$ accuracy.

Gaussian quadrature in practice

This downside of Gaussian quadrature is solving the nonlinear equations for the weights and the quadrature points.
Let us consider a very simple problem in general. Let our panel span $[0, L]$, and let us suppose we want to use two points.

$$I_G = L\left[A_1 f(\alpha_1 L) + A_2 f(\alpha_2 L)\right].$$

We expect fourth order convergence, so we should be able to match terms for the polynomial test function

$$p(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3.$$

Gaussian quadrature in practice

So, we try Gaussian quadrature...

$$\begin{aligned} I_G &= L\left[A_1 p(\alpha_1 L) + A_2 p(\alpha_2 L)\right] \\ &= c_0 L(A_1 + A_2) + c_1 L^2(A_1\alpha_1 + A_2\alpha_2) + \\ &\qquad c_2 L^3(A_1\alpha_1^2 + A_2\alpha_2^2) + c_3 L^4(A_1\alpha_1^3 + A_2\alpha_2^3) \end{aligned}$$

...when we know the exact integral is

$$I = c_0 L + c_1 \frac{L^2}{2} + c_2 \frac{L^3}{3} + c_3 \frac{L^4}{4}.$$

Gaussian quadrature in practice

So, we arrive at the nonlinear system of equations:

$$\begin{aligned} A_1 + A_2 &= 1 \\ A_1\alpha_1 + A_2\alpha_2 &= \frac{1}{2} \\ A_1\alpha_1^2 + A_2\alpha_2^2 &= \frac{1}{3} \\ A_1\alpha_1^3 + A_2\alpha_2^3 &= \frac{1}{4} \end{aligned}$$

What is solution? Ask `maple`!

In fact, it makes more sense to probe the symmetric interval $[-L/2, L/2]$. By symmetric, we expect $\alpha_1 = -\alpha_2$ and $A_1 = A_2$. We expect this method to be correct for polynomials of degree 3, so we should match up terms for the test polynomial

$$p(x) = c_0 + c_1 x + \frac{c_2}{2}x^2 + \frac{c_3}{6}x^3.$$